

AqualInsight

AI-Powered Freshwater Quality Report

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1. Executive Summary

This report presents an analysis of freshwater quality data from 5 monitoring stations over 2024-01-01 to 2024-12-30. A total of 1,825 records were analyzed using the Weighted Arithmetic Water Quality Index (WQI) and Isolation Forest anomaly detection.

2. Key Metrics

Total Records	1,825
Monitoring Stations	5
Mean WQI	74.87
Minimum WQI	23.26
Maximum WQI	99.14
Anomalies Detected	92
Active Alerts (WQI < 50)	113

3. WQI Categories Distribution

- Good: 1070 (58.6%)
- Fair: 408 (22.4%)
- Excellent: 234 (12.8%)
- Poor: 110 (6.0%)
- Very Poor: 3 (0.2%)

4. Station Performance

Station	Mean WQI	Min WQI	Max WQI	Anomalies
Station_A	70.58	27.47	96.69	25
Station_B	80.34	24.51	99.14	18
Station_C	84.10	25.96	98.68	8
Station_D	67.82	23.26	92.13	15
Station_E	71.50	28.03	94.07	26

5. Critical Alerts

113 alerts detected (WQI < 50). Top 10 listed below:

Date	Station	WQI	pH	DO (mg/L)
2024-01-07	Station_A	42.64	5.48	5.09
2024-01-18	Station_A	32.84	8.80	11.46
2024-02-06	Station_A	33.09	9.01	12.96
2024-02-09	Station_A	31.38	8.72	12.49
2024-02-11	Station_A	46.55	8.51	6.46
2024-02-23	Station_A	33.21	8.81	13.13
2024-03-07	Station_A	44.84	5.34	6.25
2024-03-15	Station_A	47.43	6.12	10.16
2024-03-30	Station_A	48.47	6.64	10.09
2024-04-03	Station_A	48.75	6.81	10.20

6. Recommendations

1. Monitor stations with consistently low WQI values (< 50) on a daily basis.
2. Investigate the root causes of detected anomalies (Industrial discharge, agricultural runoff).
3. Implement real-time IoT sensors at high-risk stations for continuous monitoring.
4. Establish a rapid response protocol for critical alerts to protect public health.
5. Conduct regular water treatment audits to maintain quality standards.

